



Father Involvement and Child Development: A Prospective Study of Syrian Refugee Families

Kristin Hadfield¹ · Majd Al-Soleiti^{2,3} · Rana Dajani⁴ · Isabelle Mareschal⁵ · Catherine Panter-Brick^{6,7,8} 

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Abstract

In refugee contexts, relatively little is known about men's child- and family-directed behaviours and even less about the impacts of father involvement. We examine father and mother reports of levels of father involvement, and their associations with family functioning and child development during forced displacement. In 2021–22, we prospectively collected data from Syrian refugee families with 4–8 year old children in Jordan, at two time points (father-mother-child triads, $n = 160$ at T1, $n = 105$ at T2). Syrian fathers viewed themselves as highly involved with their children, but mothers disagreed: fathers rated themselves as 13% more involved with their children than mothers reported them to be. Levels of father involvement were largely unrelated to child outcomes. However, spousal disagreement about father involvement was negatively associated with relationship quality, mother mental health, and child social emotional learning (SEL). In the most discordant families, child SEL was more than a standard deviation (43%) below levels observed in the most concordant families. This study builds the evidence base for the impacts of father involvement in refugee families and deepens understanding of father-mother reporting discordance. In terms of impacts, there is an important distinction between *levels of father involvement* and *levels of father-mother agreement* about father involvement. Helping parents negotiate caregiving interactions and responsive parenting may benefit child mental health and social emotional learning.

Keywords Child development · Family functioning · Fathers · Mental health · Parenting · Refugees

Highlights

- Syrian refugee fathers viewed themselves as highly involved with their 4–8 year-old children; their wives did not.
- Levels of father involvement were largely unrelated to child outcomes.
- Discordance about father involvement was negatively associated with family relationships, mother mental health, and child SEL.
- In the most discordant families, child SEL was 43% below levels observed in the most concordant families.
- Helping parents negotiate fathering-related expectations may benefit children's mental health and SEL.

These authors contributed equally: Kristin Hadfield, Majd Al-Soleiti

✉ Catherine Panter-Brick
Catherine.Panter-Brick@yale.edu

¹ Trinity Centre for Global Health, School of Psychology, Trinity College Dublin, Dublin, Ireland

² Taghyeer Foundation, Amman, Jordan

³ Department of Psychiatry, Mayo Clinic, Rochester, USA

⁴ Faculty of Science, The Hashemite University, Zarqa, Jordan

⁵ School of Behavioural and Biological Sciences, Queen Mary University of London, London, UK

⁶ Jackson School of Global Affairs, Yale University; Horchow Hall, New Haven, USA

⁷ Department of Anthropology, Yale University, Yale, USA

⁸ Conflict, Resilience, and Health Program, Yale University, Yale, USA

Relatively little is known about men's child and family-directed behaviours in contexts of war and forced displacement and, specifically, whether levels of father involvement can impact child health and development outcomes in refugee populations. Yet engaging with men, in their role as fathers, is important for improving responsive parenting, family support and cohesiveness, mental health, and healthy development (Dingus et al., 2022; Leckman et al., 2021; Martin et al., 2022; McLean & Panter-Brick, 2022; Miller et al., 2023). Understanding disagreements in how spouses view father involvement may also shed light on family relationships, mental health, and child development. In this paper, we examine father involvement in Syrian refugee families (father, mother, child triads) in Jordan. We specifically focus on the extent to which father involvement is associated with children's mental health and social emotional learning. We also examine how discrepancies between mothers and fathers in terms of the level of father involvement are associated with these outcomes.

Father involvement can have a pivotal role in multiple areas of child development. It includes engagement, availability, and responsibility (Lamb et al., 1985), as well as indirect care and process responsibility (Pleck, 2012). Direct father-child interaction fosters social emotional learning and the establishment of secure attachments (Amodia-Bidakowska et al., 2020; Barker et al., 2017; Lamb, 2010; Piskernik & Ahnert, 2019; Yoon et al., 2021). Fathering involvement with the child's mother contributes to spousal relationship quality and to the familial environment, both of which are consequential for child development (Cummings et al., 2004; Khraisha et al., 2024). Involvement as a father in the community models constructive social engagement and is important to the social networks of both the parents and the child (Hein et al., 2020; McLean & Panter-Brick). Despite its importance, evidence for the impacts of father involvement on family functioning and child development primarily comes from high-income countries, with a dearth of research in low- and middle-income countries (Garcia et al., 2022; Schoppe-Sullivan and Fagan 2020). Yet, fathering in these contexts may be different (Volling & Cabrera, 2019) and it is important to build evidence for the impacts of father involvement in contexts of violence and social adversity, where children face disadvantage at multiple levels (Jeong et al., 2018; Hein et al., 2020; Ndiaye and Basil Tandika, 2022; Shafer & Easton, 2021; Feldman, 2023). In particular, there is a dearth of evidence from refugee contexts, despite the scale of ongoing refugee crises, whereby 1% of people worldwide are forcibly displaced, and most displaced people are children (UNHCR, 2022a).

Refugee children face multiple challenges to their development (Arakelyan & Ager, 2021; Flanagan et al., 2020; Miller & Rasmussen, 2017; Panter-Brick et al.,

2018a, 2020). Trauma, forced displacement, and resettlement impact children's health and social competence across the life course directly, as well as indirectly through compromised caregiving (Bryant et al., 2018; Beni Yonis et al., 2021; Eltanamly et al., 2021; Fazel et al., 2011; Flanagan et al., 2020; Frounfelker et al., 2020; MacDermid Wadsworth, 2010; Reed et al., 2011; van Ee et al., 2013). Consequently, displaced children have poorer mental health than their non-refugee peers (Frounfelker et al., 2020; Hodes & Vostanis, 2019) and their social emotional learning may be negatively affected (Lee, 2019; Michalek et al., 2023; Sullivan & Simonson, 2016). Knowing the importance of father involvement, there are calls to make dedicated efforts to consider fathers' roles in refugee children's development, beyond the habitual focus on mother-child dyads (Feldman, 2023; McLean & Panter-Brick, 2022).

Research in crisis-affected and low-resource settings rarely includes fathers' reports of their own roles. For example, in Lebanon, an intervention study of refugee and low-income families used mothers' reports to gather data about fathers' involvement with their young children, spouses, and community (Hein et al., 2020). Similarly, a study of refugee families of 8–16-year-olds in Lebanon described parenting relationships using data from a sample comprised almost exclusively of mothers (Popham et al., 2022). By contrast, Miller et al.'s (2023) caregiver support intervention evaluation of Syrian refugee and Lebanese non-refugee families of 3–12-year-old children in Lebanon was able to recruit both male and female caregivers.

In both displaced and non-displaced families, father-child relationships are, more often than not, studied through the lens of mother-reported data. This is problematic. The handful of studies that explicitly compare father and mother reports of father involvement have noted substantive discrepancies, with mothers viewing fathers as less involved than fathers view themselves. For example, Coley & Morris (2002) examined couple-level characteristics that predicted discrepancy in father involvement reports in matched pairs of fathers and mothers from the *Three City Study* in the United States (Coley & Morris 2002). Fathers consistently reported higher levels of involvement relative to what mothers reported about fathers, although spousal reports were highly correlated. Similarly, Mikelson (2008) used the American *Fragile Families and Child Wellbeing* cohort to investigate the demographic and social factors predicting father-mother discrepancy in views of father involvement. Fathers indicated spending 17.6% more time – amounting to an additional 0.6 days per week – than mothers had reported fathers spent with their children. These and other articles (e.g. Hernandez & Coley, 2007; Matte-Gagné et al., 2021) discuss whether fathers or mothers are more reliable respondents, and the possibility that fathers inflate and

mothers deflate fathers' child-directed involvement. Despite known cross-cultural differences pertaining to fatherhood (Seward & Stanley-Stevens, 2014; Yeh et al., 2021), there are to-date no systematic comparisons of father- and mother-reports of child-directed father involvement in countries outside of the United States, Sweden (Wängqvist et al., 2022), or Japan (Terui et al., 2021). Additionally, there is a dearth of research on father-mother disagreement about how fathering involvement relates to other aspects of family function. Yet, based on previous work on spousal relationships and parenting, we may expect that a spousal discrepancy in understanding father involvement might be indicative of a lack of effective communication and a divergence in parenting expectations, that could lead to stress and insecurity (e.g. Belsky, 1984; Diniz et al., 2021).

To address these gaps, we collected data on Syrian refugee families in Jordan, at two timepoints during 2021–22. The study featured both father- and mother-reports of father involvement, allowing for explicit comparison of spousal data agreement or disagreement. Mothers and children, but not fathers, were engaged in a randomised controlled trial evaluating a reading intervention, known as *We Love Reading* (Dajani, 2017; Mareschal et al., 2022), implemented to boost child literacy and attitudes towards reading. To date, Jordan is the second-largest host of Syrian refugees per capita, after Lebanon. Some 656,762 Syrians are registered as refugees or asylum-seekers in Jordan, 80% of whom live outside of refugee camps (UNHCR, 2023); the remainder live in refugee camps, the largest of which is Zaatari. As with most refugees worldwide, Syrians displaced in Jordan live in a protracted refugee situation, and are limited in terms of work opportunities (UNHCR, 2022a) as well as access to healthcare (UNHCR, 2022b) and schooling (Christophersen, 2015). These refugees face substantial challenges: 85% live below the poverty line, 12% have a serious medical condition, and, among children, 69% of those who attend school attend a double-shifted school, with classes in the morning or afternoon only (Ministry of Education, 2018; UNHCR, 2023; UNICEF Jordan, 2019).

Our main research questions were twofold:

(RQ1) To what extent are fathers engaged with their families and communities? How are levels of father involvement associated with child mental health and social emotional learning in this refugee sample?

(RQ2) Are levels of spousal disagreement about father involvement associated with (i) family functioning and relationship quality; (ii) father, mother, and child mental health; and (iii) child social emotional learning?

In a series of pre-registered analyses, we examined both cross-sectional and prospective data in order to establish associations (at T1 and T2) and directional effects (T1 → T2).

For RQ1, we hypothesised that Syrian refugee men would report high levels of family engagement with their children, spouses, and communities, and that fathers would report higher levels of family engagement, as compared to mothers' ratings of the fathers. Given a dearth of research on impacts of father involvement in refugee families, we did not have directional hypotheses for how father involvement would impact child mental health or social emotional learning. For RQ2, we hypothesised that greater spousal discordance in reports of father engagement with the family would be associated with worse family functioning, and poorer mother-father and mother-child relationship quality. Given a lack of research on this topic, we did not have directional hypotheses for how spousal disagreement about father involvement would be associated with father, mother, or child mental health, or with child social emotional learning; instead, we conducted exploratory analyses.

Methods

Procedures

In this paper, we analyse data on father-mother-child triads, focusing on levels and impacts of father involvement; we do not examine outcomes of the randomised trial (Hadfield et al., 2024). The study was approved by the ethics committees of Yale University (1502015359) and Trinity College Dublin (01E/2020/10) as well as by the Prime Minister's Office (23849/4/14/1) in Jordan. Syrian families were recruited through community-based organizations that serve low-resource Jordanian populations as well as refugees, in Amman and Zaatari. Field researchers ($n = 9$) were native Arabic speakers with university or professional qualifications, experienced in data collection. Six female researchers visited mother-child dyads in their home, two male researchers interviewed the fathers by phone, and one project manager coordinated all field activities. Fathers were contacted only if mothers explicitly consented to a spousal interview. Survey timelines differed slightly for father and mother-child data collection (Fig. S1) because funding to interview the fathers was received separately, after the start of the mother-child trial. Fathers were interviewed at two time-points: March–July 2021 (T1) and August–January 2021–22 (T2); mothers and children were interviewed during February–May 2021 (T1) and May–August 2021 (T2), corresponding to baseline and post-intervention trial periods. Mothers and fathers provided written informed consent for themselves. Mothers provided written informed consent for the study child and then the children were asked verbally if they would take part.

Participants

The trial followed 322 mothers and their 4–8 year-old children (50.0% female, see Hadfield et al., 2022, for more information). At T1, we contacted 222 fathers from these families and collected data from 160; 62 fathers were not interviewed, being unreachable by phone ($n = 44$), or unwilling to participate ($n = 18$). In 103 cases, we did not seek to collect father data: we excluded 73 families whose mother and child data had already been collected, and men who were divorced ($n = 10$), unavailable ($n = 8$), deceased ($n = 3$), travelling ($n = 2$), or missing ($n = 1$); we also excluded families for whom mothers did not give consent to father participation ($n = 6$). At T2, which was approximately 4 months later, we sampled 105 triads: 53 of the fathers from T1 could not be reached, while 2 refused to participate; all mothers and children from these 105 families participated at T2. Our total sample thus included 160 father-mother-child triads at T1 and 105 triads at T2. The sample retention rate was 65.63%. Overall, only 6 mothers and 20 fathers refused study participation.

We examined two potential sampling biases. With respect to inclusion at T1, there were few differences between the families with participating father-mother-child triads and families with non-participating fathers. There were no differences in mother age, study child gender, number of children in the household, or T1 mother-reported father-child, father-mother, or father-community involvement ($ps > 0.05$); however, the former had lower household wealth ($M = 7.70$ v. 8.14) and their index study child was younger ($M = 6.16$ v. 6.48 years). With respect to T1-T2 attrition bias, there were no sample differences in terms of household wealth, child age, mother age, number of children in the household, or mother-reported father involvement ($ps > 0.05$); the only difference was child gender, with families with a boy as index child less likely to be retained at T2.

Families had left Syria and arrived in Jordan between 2012 and 2014. More than two-thirds (69.4%) lived in socioeconomically deprived areas of Amman and 30.6% in Zaatari refugee camp. Fathers averaged 37.41 ($SD = 7.37$), mothers 32.61 ($SD = 7.02$), and children 6.32 ($SD = 1.18$) years of age. Fathers had an average 4.65 children ($SD = 2.02$), mothers an average 4.24 ($SD = 1.93$) children; 5.0% of fathers had more than one wife. Half the children were female.

Most fathers had left school after primary education; one in 7 fathers (13.8%) and 1 in 6 mothers (17.4%) indicated that they were not able to read. At T1, 76.9% of fathers were employed – mostly part-time (69.9%), rather than full-time (30.1%), with 23.1% having no job. Six months later (T2), 67.0% were employed, most of these (71.6%) part-

time, rather than full-time (28.4%). Only 10.9% of mothers were employed at T1. Fathers' employment status was unrelated to family residence in Amman or Zaatari ($p > 0.05$, T1 and T2).

Measures

Survey tools were chosen by an interdisciplinary panel, comprised of local and western research team members. Where possible, we included measures previously developed and validated with displaced populations in the region; where not already in Arabic, they were translated into Arabic, back-translated into English, iteratively reviewed, and final versions agreed upon through consensus discussion (Valdez et al., 2021). Tools were piloted with mother-child dyads ($n = 10$) and with fathers ($n = 8$) before both timepoints. Mother-child data collection took approximately one hour (one fieldworker interviewed the mother; another assessed the child at the same time) and father data collection took 45 min. Survey questions were read aloud, and responses inputted into KoboToolbox by field researchers. Participants were compensated for their time with a small payment.

Data on father involvement were collected from both spouses; other family-level data were collected from one spouse only, to avoid burdening respondents with lengthy interviews. The T1 survey was implemented at the height of COVID restrictions in Jordan. We included more measures at T2, lengthening data collection to get a richer picture of parenting behaviours, mental health, and child SEL (Fig. S1). Table 1 presents the psychometric properties of all included measures (means, standard deviations, internal reliability).

Father Involvement

Father involvement (father and mother reports, T1 and T2) We used the Father Involvement Questionnaire (FIQ) developed for use with refugee and low-resource communities in Lebanon (Hein et al., 2020), a 20-item measure with three subscales assessing father-child, father-mother, and father-community involvement. We adapted the FIQ to be used as a father self-report, in addition to using the original version which asks the mother to report about her spouse. The FIQ includes items such as “I am knowledgeable about my child(ren)’s education” (father-child subscale), “I treat my spouse well in front of our child(ren)” (father-mother subscale), and “I spend time with other fathers in the community” (father-community subscale). Items have a 4-point Likert-scale response ranging from strongly disagree (1) to strongly agree (4), with higher scores indicating greater involvement.

Table 1 Means, standard deviations, and Cronbach's alphas for study measures, by father and mother respondent

	Fathers						Mothers					
	T1			T2			T1			T2		
	<i>M</i>	<i>SD</i>	α	<i>M</i>	<i>SD</i>	α	<i>M</i>	<i>SD</i>	α	<i>M</i>	<i>SD</i>	α
<i>Father involvement</i>												
Father-child involvement (FIQ)	30.25	3.79	0.63	30.32	3.50	0.66	26.64	4.96	0.82	26.54	4.66	0.85
Father-mother involvement (FIQ)	22.44	1.86	0.67	23.11	1.53	0.66	19.42	3.62	0.88	19.59	3.13	0.84
Father-community involvement (FIQ)	13.22	3.66	0.73	13.62	3.43	0.64	13.60	2.55	0.76	13.60	2.55	0.74
<i>Family functioning and relationship quality</i>												
Family functioning (GF6+)	21.99	2.23	0.72	22.53	2.21	0.74						
Spousal relationship quality (DAS)							22.67	6.36	0.74	26.35	8.56	0.83
Mother-child closeness (Pianta)							33.01	2.43	0.42	32.76	2.44	0.65
Mother-child conflict (Pianta)							23.43	6.54	0.65	22.88	6.65	0.74
Parenting self-agency (PSA)										31.19	3.87	0.76
Mother-child positive parenting (APQ)										13.90	1.45	0.61
Mother-child inconsistent discipline (APQ)										10.26	2.61	0.46
Mother-child poor supervision (APQ)										4.45	2.16	0.60
<i>Mental health and wellbeing</i>												
Wellbeing (WEMWBS)				57.16	9.80	0.87				49.58	9.78	0.85
Depressive symptoms (CESD)				9.15	6.63	0.86				14.02	6.30	0.79
Anxiety symptoms (DASS)				2.51	3.30	0.80				6.65	4.61	0.78
Attentive problems (PSC-17)										5.12	2.32	0.77
Internalising problems (PSC-17)										2.93	1.90	0.57
Externalising problems (PSC-17)										4.95	3.20	0.59

Family Functioning and Relationship Quality

Family functioning (father report, T1 and T2) We implemented the 6-item General Functioning Subscale (GF6+) of the McMaster Family Assessment Device (Botelho et al., 2015). Items include “In times of crisis we can turn to each other for support.” Items have a 4-point Likert-scale response ranging from strongly disagree (1) to strongly agree (4), with higher scores indicating healthier family functioning.

Spousal relationship quality (mother report, T1 and T2)

We used the 8-item Arabic Dyadic Adjustment Scale (DAS, Elanbari, 2015; Spanier, 1989). Items include, “How often do you and your husband ‘get on each other’s nerves’?”, with higher scores on the 6-point Likert scale indicating a better quality relationship.

Mother-child closeness and conflict (mother report, T1 and T2) The 15-item Pianta Child-Parent Relationship Scale – Short Form (Pianta, 1992) consists of two subscales: closeness (7 items) and conflict (8 items). Items include: “I share an affectionate, warm relationship with my child” (closeness) and “My child and I always seem to be

struggling with each other” (conflict). This measure has a 5-point Likert scale ranging from 1 (definitely does not apply) to 5 (definitely applies), with higher scores indicating greater closeness or greater conflict.

Parenting self-agency and mother parenting behaviours

(mother report, T2) The Parental Self-Agency Measure (PSA, Dumka et al., 1996) includes 5 items, such as “I can solve most problems between my child and me.” The Likert scale ranges from rarely (1) to always (7), with higher scores indicating greater self-agency as a mother. The 9-item Alabama Parenting Questionnaire – Short Form (APQ, Elgar et al., 2007) has three subscales assessing parenting behaviours: positive parenting, inconsistent discipline, and poor supervision (three items each). Items include: “You praise your child if they behave well” (positive parenting), “Your child talks you out of being punished after they have done something wrong” (inconsistent discipline), and “Your child stays out in the evening after the time they are supposed to be home” (poor supervision). Responses on the 5-point Likert-scale range from never (1) to always (5), with higher scores indicating more positive parenting, more inconsistent discipline, and poorer supervision.

Father, Mother, Child Mental Health

Wellbeing (father and mother report, T2) The 14-item Warwick-Edinburgh Mental Wellbeing Scale (WEMWBS) assesses subjective wellbeing, with higher scores indicating greater wellbeing. Items include: “I’ve been feeling optimistic about the future”, with responses ranging from 1 (none of the time) to 5 (all of the time). This questionnaire was implemented half-way through the data collection with mothers (to replace a trauma exposure checklist, which mothers disliked completing), and with all fathers ($n = 79$ mothers, $n = 93$ fathers).

Depressive symptoms (father and mother report, T2) The 10-item Centre for Epidemiological Studies – Depression Scale (CES-D, Radloff, 1977) assessed depressive symptoms on a 4-point Likert scale, ranging from 0 (rarely, or never) to 3 (most, or almost all the time). How respondents felt or behaved over the past month was elicited with statements such as “I felt that everything I did was an effort”. Higher scores are indicative of greater symptom severity.

Anxiety symptoms (father and mother report, T2) The anxiety subscale of the short form Depression, Anxiety, Stress Scale (DASS-21, Henry & Crawford, 2005) measures symptoms of anxiety (7 items). The DASS-A is assessed using a 4-point Likert-scale ranging from 0 (did not apply to me - never) to 3 (applied to me very much, or most of the time – almost always). Reports over past month are elicited with items such as “I felt scared without any good reason.” Higher scores indicate more anxiety symptoms.

Child mental health (mother report, T2) The 17-item Paediatric Symptom Checklist (PSC-17, Gardner et al., 1999) measures child attentive, internalising, and externalising problems (three subscales, 5-item each). It uses a 3-point Likert scale, with options of never (0), sometimes (1), and often (2). Higher scores indicate more attentive, internalising and externalising problems.

Child social emotional learning

Child social emotional learning (child direct assessment, T2) Children completed the Holistic Assessment of Learning and Development Outcomes (HALDO; D’Sa, Krupar, & Westrope, 2019). The HALDO was developed to be administered to refugee children aged 4–12 years in low- and middle-income country contexts, and has been used previously with refugees in Lebanon (e.g. D’Sa et al., 2019). This SEL assessment consists of two components: self-concept and empathy. Self-concept is the understanding of and ability to express personal

preferences, feelings, thoughts, and abilities; items include, “Please tell me the name of one person who takes care of you at home” and “Please tell me one thing that you hope/wish will happen in your life in the future.” To assess empathy, children were shown a picture of a similar-aged child crying and were asked, “How do you think this child feels?”, followed by “What would you do to help them feel better?” Responses were scored as 0 (incorrect) or 1 (correct), then summed across items to create the SEL score ($M = 8.44$, $SD = 3.44$, range: 0–16).

Contextual Information

Education (father and mother report, T1), household wealth (mother report, T1), and employment (father report, T1 and T2) Participants reported education levels as having no schooling, grade 1–6, grade 7–12, or university education. For socioeconomic status, we implemented the Household Index, a checklist developed for use in humanitarian settings (Panter-Brick et al., 2018b). Respondents were asked whether they had the following items: TV, satellite, smartphone, car, refrigerator, computer, oven with gas, bedframe, washing machine, heater, fan, and water heater. Higher scores indicate greater household wealth. Parents were asked if they worked (0 = no, 1 = yes) and if so, if it was full-time or part-time.

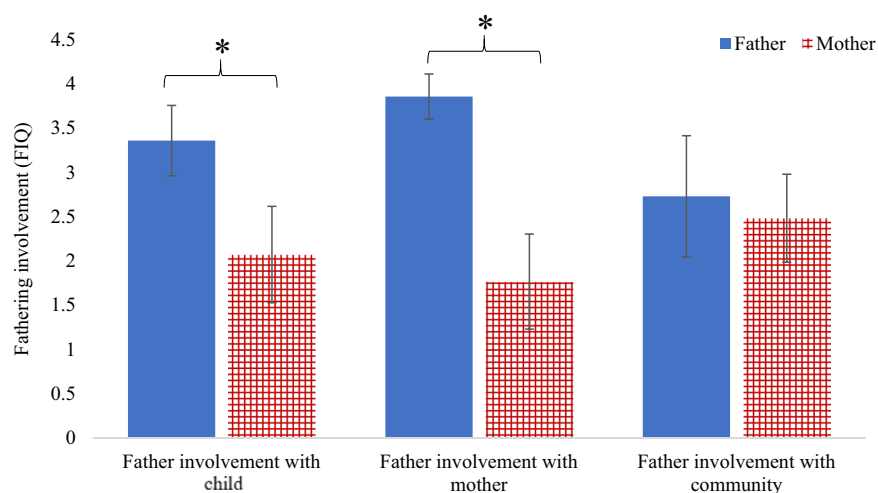
Statistical Analyses

All measures are available online, in both English and Arabic. All T1, T2, and T1 → T2 analyses were [pre-registered](#) on [OSF](#). The p-threshold for all analyses is set at 0.05.

To better understand levels of father involvement and differences in how fathers and mothers view this, we conducted paired samples t-tests to compare father and mother FIQ scores at T1 and T2. We then calculated absolute levels of spousal disagreement (‘discordance’) in reports of father involvement, based on father and mother FIQ scores. To address RQ1, we ran linear regressions with father- and mother-reported FIQ scores predicting child outcomes (attentive problems, internalizing problems, externalizing problems, and SEL), controlling for household wealth and father education. We did this first cross-sectionally (testing associations at T2) and then prospectively, examining changes over time (T1 predictors → T2 outcomes). Both analyses showed largely the same pattern of results and so below we present the prospective analyses (how T1 father involvement predicts T2 outcomes), with the T2 cross-sectional analyses in Table S1.

To address RQ2, we used linear regressions to understand how FIQ spousal disagreement was associated with (i) family functioning and relationship quality, (ii) father,

Fig. 1 Comparison of father involvement data, as reported by fathers (blue) and mothers (red). *Note:* This figure shows T2 mean scores, rather than the sum scores reported in text, to facilitate comparisons across different subscales. $*p < 0.05$ difference between father and mother reports



mother, and child mental health, and (iii) child SEL, controlling for household wealth and father employment. We did this first cross sectionally, for T1 and T2 data separately. We then looked for directional effects by using T1 spousal discordance data to predict T2 outcomes. We present below the prospective analyses (how T1 spousal discordance predicts T2 outcomes). We report cross-sectional analyses in supplementary tables: family functioning and relationship quality in Table S2; father, mother, and child mental health in Table S3; and child SEL in Table S4. As can be seen in Table S2, the pattern of results for the cross-sectional analyses at T1 and at T2 is broadly the same.

In analyses for RQ1 and RQ2, we included two covariates: household wealth and father education. Their inclusion is based on research showing that paternal education and levels of poverty are related to family function and spousal relationship quality (e.g. Cabrera et al., 2014; Conger et al., 2010; Dakin & Wampler, 2008). While other control variables are potentially relevant, we included only these two (as specified pre-registration), given our sample size, in order to limit the chance of Type II error and conserve power.

We have included two covariates in our analyses: household wealth and father education that we indicated in our pre-registration would be the only controls. The inclusion of these covariates is based on research that paternal education and levels of poverty are related to spousal relationship quality and family function (e.g. Cabrera et al., 2014; Conger et al., 2010; Dakin & Wampler, 2008). While there are other control variables that may be relevant, we have chosen to limit these given our sample size in order to conserve power. One potential control which we did not include was whether children were taking part in the We Love Reading shared book reading intervention. The primary outcomes for that intervention were attitudes toward reading and literacy and thus non-relevant to our analyses

here. In another paper, we evaluate the effects of this intervention on family function – specifically mother-child closeness, mother-child conflict, spousal relationship quality, and father involvement; none of these were meaningfully impacted by the intervention (Hadfield et al., 2024) and thus we have not controlled for intervention group (treatment, control) in any analyses.

Results

Reports of Father Involvement

Fathers reported levels of involvement with children and spouses that were higher than those reported by mothers (Fig. 1). Spouses disagreed about father-child involvement ($t(138) = 7.78, p < 0.001$ at T1, $t(99) = 7.88, p < 0.001$ at T2) and about father-mother ($t(146) = 9.05, p < 0.001$ at T1, $t(99) = 10.52, p < 0.001$ at T2) involvement, but agreed about levels of father-community involvement ($p > 0.05$, T1 and T2). For the father-child subscale, at T2 fathers reported being 13.3% more involved than mothers reported them to be (12.7% at T1), and for the father-mother subscale, they reported being 16.5% more involved than mothers did at T2 (14.4% at T1). FIQ scores and patterns of spousal discordance were largely constant over time, except for small T1-T2 increases in father-reported father-mother involvement and in mother-reported father-community involvement (Table S5). The distribution of father involvement scores, as reported by fathers and mothers, is shown in Fig. S2.

Impacts of Father Involvement (RQ1)

Father reports of their involvement were not predictive of child attentive problems, internalising problems, or SEL ($p > 0.05$, Table S1). However, where fathers reported

Table 2 Prospective associations between spousal discordance about father involvement and family functioning and relationship quality

Outcome variables at T2	Spousal discordance in reports of father involvement at T1					
	Panel A Father-child involvement		Panel B Father-mother involvement		Panel C Father-community involvement	
	R^2	β	R^2	β	R^2	β
<i>Father-report</i>						
Family functioning (GF6 +)	0.04	<0.01	0.04	<0.01	0.07	−0.14
<i>Mother-report</i>						
Spousal relationship quality (DAS)	0.14	−0.36**	0.21	−0.45**	0.01	0.01
Mother-child closeness (Pianta)	<0.01	−0.06	0.02	−0.14	<0.01	0.01
Mother-child conflict (Pianta)	0.04	0.14	0.04	0.19*	0.02	−0.14
Parenting self-agency (PSA)	0.01	−0.01	0.06	−0.22	0.03	0.12
Maternal positive parenting (APQ)	0.10	−0.24*	0.05	−0.16	0.02	−0.02
Maternal inconsistent discipline (APQ)	0.01	0.03	0.01	<0.01	0.02	0.05
Maternal poor supervision (APQ)	0.04	0.13	0.02	0.08	0.06	−0.21*

FIQ Father Involvement Questionnaire, *GF6+* General Functioning subscale, *DAS* Dyadic Adjustment Scale, *PSA* parenting self-agency scale, *APQ*, Alabama Parenting Questionnaire

Separate linear regression analyses were run for subscale of spousal discordance (assessed at T1) and for each outcome variable as assessed at T2, controlling for father employment and household wealth. Significance level, * $p < 0.05$, ** $p < 0.001$

greater father-child involvement, children had fewer externalising problems ($B = -0.15$, $p = 0.039$, 95% CI: $-0.29, -0.01$). Similarly, mother reports of father involvement did not predict child mental health or SEL ($p > 0.05$, Table S1). Almost all analyses are non-significant: levels of father involvement, as reported by either fathers or mothers, were largely unassociated with child mental health or social emotional learning.

Levels of Spousal Discordance

Levels of spousal discordance were positively correlated for father-child and father-mother subscales ($r = 0.51$, $p < 0.001$), but uncorrelated with the father-community subscale ($r = 0.03$ and $r = 0.05$, respectively, $p > 0.05$). That is, spouses who showed more disagreement about how the father was involved with the child, also had more disagreement about how the father was involved with the mother, but did not disagree about how much the father was involved in the community. Levels of spousal discordance were unrelated to family residence (Amman versus Zaatari, $p > 0.05$, T1 and T2) or household wealth ($p > 0.05$, T1 and T2).

Impacts of Spousal Discordance (RQ2)

Associations with family functioning and relationship quality

Levels of T1 spousal FIQ discordance were not associated with T2 father-reported family functioning; however, they

were associated with mothers' reports of their spousal relationships, mother-child relationship quality, and mother parenting (Tables 2, S2). As shown by panel A of Table 2, regression models of spousal discordance in reports of father-child involvement show that when fathers and mothers disagreed about how involved fathers were with the child, mothers reported poorer spousal relationships ($\beta = -0.36$, $p < 0.001$) and less positive parenting ($\beta = -0.24$, $p = 0.006$) (Panel A of Table 2). Models explained 14% of the variance in spousal relationship quality and 10% of variance in maternal positive parenting, respectively.

Panel B shows regression models of spousal discordance in father-mother involvement: when fathers and mothers disagreed about his involvement with her, mothers felt their spousal relationship was of lower quality ($\beta = -0.45$, $p < 0.001$), and their relationship with the child more conflicted ($\beta = 0.19$, $p = 0.027$); they also felt less self-agency as a parent ($\beta = -0.22$, $p = 0.010$). Models explained a large proportion (21%) of the variance in spousal relationship quality, and a small proportion of the variance in mother-child conflict (4%) and mother parenting self-agency (6%).

Panel C shows that discordant FIQ scores pertaining to father-community involvement were unrelated to family functioning or to the quality of spousal and mother-child relationship, except poor supervision: when spouses were more discordant about father involvement with the community, mothers reported poorer supervision of their children ($\beta = -0.21$, $p = 0.015$). This was a small effect;

Table 3 Prospective associations between discordance about father involvement and mental health and child social emotional learning

Outcome variables at T2	Spousal discordance in reports of father involvement at T1					
	Panel A Father-child involvement		Panel B Father-mother involvement		Panel C Father-community involvement	
	R^2	β	R^2	β	R^2	β
<i>Father mental health</i>						
Wellbeing (WEMWBS)	0.05	0.22	0.02	0.07	0.02	0.10
Depressive symptoms (CESD)	0.01	−0.03	0.01	0.06	<0.01	0.03
Anxiety symptoms (DASS)	0.01	−0.02	0.01	−0.07	0.02	0.10
<i>Mother mental health</i>						
Wellbeing (WEMWBS)	0.06	−0.18	0.08	−0.24*	0.02	0.02
Depressive symptoms (CESD)	0.03	0.12	0.05	0.21*	0.04	−0.15
Anxiety symptoms (DASS)	0.01	−0.04	0.02	0.06	0.01	−0.03
<i>Child mental health</i>						
Attentive problems (PSC)	0.03	0.14	0.02	0.10	0.03	−0.09
Externalizing problems (PSC)	0.01	−0.01	0.02	0.07	0.02	−0.10
Internalizing problems (PSC)	0.01	0.06	0.01	0.10	<0.01	−0.05
<i>Child social emotional learning</i>						
Social emotional learning (HALDO)	0.06	−0.12	0.07	−0.21*	0.03	−0.02

FIQ Father Involvement Questionnaire, *WEMWBS* Warwick Edinburgh Mental Wellbeing Scale, *CES-D* Centre for Epidemiological Studies – Depression Scale, *DASS* anxiety subscale of the short form Depression, Anxiety, Stress Scale, *PSC* Paediatric Symptom Checklist (mother-reported), *HALDO* Holistic Assessment of Learning and Development Outcomes (direct assessment)

Separate linear regression analyses were run for each subscale of spousal discordance and each outcome variable, controlling for father employment and household wealth. Child attentive, externalizing, and internalizing problems were mother-reported; child social emotional learning (SEL) was by direct assessment. Significance level, * $p < 0.05$

the model explained 6% of the variance in mother supervision of her children. The same broad pattern of results was observed for cross-sectional T2 data, with discordance about father-community involvement largely unrelated to family functioning or relationship quality (Table S3).

Associations with father, mother, and child mental health

Levels of spousal FIQ discordance at T1 were associated with the mothers' wellbeing and depressive symptoms at T2, but not with father or child mental health (Table 3). Specifically, where spouses disagreed about father-mother involvement (Panel B), mothers reported lower wellbeing ($\beta = -0.24$, $p = 0.042$) and more symptoms of depression ($\beta = 0.21$, $p = 0.013$). These effects were small or medium in size, with discordance about father-mother involvement accounting for 8% of the variance in mother wellbeing and 5% in mother depressive symptoms.

Associations with child social emotional learning

Lastly, we examined the extent to which father-mother disagreement regarding father involvement at T1 predicted

children's SEL at T2, as directly assessed by the HALDO. When fathers and mothers disagreed more about father-mother involvement, children had lower SEL scores (Panel B, $\beta = -0.21$, $p = 0.012$). These effects were small-to-medium in size, with discordance about father-mother involvement accounting for 7% of the variance in child SEL. Spousal discordance about father-child or father-community fathering involvement was not associated with child SEL.

Discussion

In this study, we prospectively followed a cohort of Syrian refugee families, assessing triads of fathers, mothers, and children at two timepoints. We examined levels of father involvement and their associations with child outcomes (RQ1), seeking to address data scarcity pertaining to fathers in refugee and war-affected communities. We also investigated whether levels of spousal discordance in reports of father involvement were associated with family functioning, relationship quality, mental health, and child social and emotional learning (RQ2). Three main findings are highlighted below.

First, Syrian refugee fathers reported being more involved with their family (wife and child) than mothers reported them to be: fathers had 13% higher scores on the father-child involvement measure (T1 and T2), relative to what mothers reported. This is in line with research conducted in Canada (Matte-Gagné et al., 2021), the United States (Charles et al., 2018; Coley & Morris, 2002; Hernandez & Coley, 2007; Mikelson, 2008), Sweden (Wängqvist et al., 2022), and Japan (Terui et al., 2021), suggesting that either fathers over-report their involvement, mothers under-report father involvement, or both. This discrepancy is important because, as Mikelson (2008) noted, “father-mother discrepancies in reporting fathers’ involvement with their children may ... impact the findings from research that uses only mother reports to examine fathers’ involvement” (p.622). To our knowledge, ours is the only study to measure and evaluate the discordance (and impact) of father-mother views pertaining to the extent to which fathers are involved with their families in a refugee sample or in any sample from the Middle East and North Africa region, with the closest work being from Lari and Al-Emadi (2021), who compared time spent parenting in unrelated fathers and mothers in Qatar.

We implemented the Father Involvement Questionnaire, originally developed in Lebanon (Hein et al., 2020), adapting it for usage with both mother and father reports. We note that while Garcia and colleagues’ (2022) survey in Kenya did evaluate family-level impacts of father and mother views of father involvement, it employed different measures for fathers and mothers. With a view to methods, a recent program evaluation among Syrian refugees in Lebanon opted to develop a new parenting measure, to better suit the cultural context and the age range of children under study (Miller et al., 2023). Developing instruments that assess the child- and family-focused behaviours of both fathers and mothers, and are relevant to diverse cultural and socioeconomic contexts, is a key step in building the evidence for the relevance of father involvement and for evaluating interventions in this population.

Second, with the exception of father-reported father-child involvement and mother-reported father-mother involvement, all components of father involvement were unrelated to child mental health or social emotional learning. That is, levels of father involvement were by-and-large unassociated with child mental health or social emotional learning. There is an extensive body of work on the importance of father involvement for children’s optimal development (e.g. Lamb, 2000), yet we found few impacts. There are a number of possible reasons for this. Firstly, much of the research on fathering has been conducted with middle class families in high-income countries (Diniz et al., 2021); father involvement may have different meanings and outcomes among a Syrian refugee

population in Jordan. There may be different expectations for fathering which neutralise the role of father involvement or make it less relevant as compared to other key family factors. It could also be that given the many and multifaceted challenges to the children’s mental health and social emotional learning, the effects of father involvement are outweighed by the other challenges faced. This would suggest that – in displacement contexts where children are faced with substantial social, economic, and environmental challenges – intervening to address some of those challenges may be more beneficial than focusing on father involvement to improve child mental health or social emotional learning. Finally, our measure of father involvement was about involvement with all of their children, yet may fathers differ in their relationship with each child (Cabrera et al., 2014; O’Connor et al., 2006), and thus our measurement may have obscured significant associations. More research is needed to disentangle these possibilities.

Third, levels of father-mother disagreement regarding father-child, father-mother, and father-community involvement were negatively associated with child outcomes – whereas actual levels of father involvement largely were not. When parents disagreed more about the father’s involvement, mothers reported poorer spousal relationship quality, more mother-child conflict, less positive parenting, poorer child supervision, lower wellbeing, and more depressive symptoms. In the most discordant families, child SEL was more than a standard deviation (43%) below levels observed in the most concordant families; importantly, SEL was assessed directly, during home visits to the family. Research is needed to understand why this spousal discordance is associated with these negative outcomes. It is possible that discordance is indicative of poor quality spousal communication or a divergence in parenting expectations, or alternately that the discordance leads to stress and insecurity which then lead to negative family and child outcomes.

In terms of family dynamics, both Syrian and Jordanian communities uphold strong social expectations that women will work within the home, raising their young children, while men will work outside the home and provide for the family (although social norms and economic expectations are changing; Almakhamreh et al., 2022; El Kharouf et al., 2021). Even when they are out-of-work, Syrian men spend much of their time outside the house. This is a context where cultural norms and financial constraints mean that many men spend limited time with their children. It is also a context where, when fathers and mothers expressed disagreement about father involvement, there were small-to-medium increases to risks of adverse child social and emotional learning outcomes. That spousal agreement or disagreement regarding men’s involvement may impact children’s development would

make sense – yet such impacts often go unmeasured in both research and impact evaluation.

In this study, we ran directional analyses with T1 predictors of T2 outcomes, and also cross-sectional analyses with available T1 and T2 data. While there are differences in specific findings, the pattern of results in these pre-registered analyses is largely the same across all analyses and timepoints. Where we do see differences, there are more significant associations in between spousal discordance and outcomes in the cross-sectional analyses than the prospective analyses. For example, cross-sectionally, but not prospectively, mothers reported more child mental health problems when spouses were more discordant about father-child and father-mother involvement – perhaps because we had more power for analyses conducted at T1 (Table S3) or because father-mother-child data were collected closer together at T1 than T2 (Fig. S1). Nonetheless, a similar pattern of results is observed, which strengthens confidence in our findings: with respect to child and family outcomes, there seems to be an important distinction between impacts of *levels of father involvement* and impacts of *levels of father-mother agreement about father involvement*.

Our findings help deepen our understanding of the impacts of fathering involvement and mother-father reporting. They also caution against two simplifying assumptions, namely that parenting programs need to engage only with mothers to learn about father involvement, and that focusing data collection on levels of father involvement – without being attentive to the quality of father-mother communication – will give important insight into child outcomes. For conflict-affected youth, a transformative approach to supporting mental health and social emotional learning requires careful attention to culturally-relevant learning environments (Stark et al., 2021), and this includes family functioning and parenting at home as well as responsiveness in schools. Our findings point to the importance of disaggregating child-directed and mother-directed fathering involvement in research and intervention projects that are able to recruit father-mother-child triads. For instance, in Garcia et al.'s (2022) study in rural Kenya, men's household-directed behaviours – through increased father support of the mother – led to small but significant improvements in children's development outcomes (age 16–34 months), while men's child-directed behaviours did not. By contrast, a large-scale study in the UK found that positive, child-directed behaviours from fathers can reduce the risks of adverse child mental health outcomes (age 2–21 months), especially in families where parents themselves report depressive symptoms (Martin et al., 2022). By implication, research will be most robust when evaluating different dimensions of child-parent engagement, including parenting activities within and outside the home. Moreover, parenting interventions may be most effective when

incorporating aspects of father-mother communication pertaining to responsive parenting. Understanding how to manage parenting expectations is important – our findings suggest that disagreements may affect parental mental health and the quality of parenting, with knock-on effects for children's learning and wellbeing.

This study has a number of limitations. First, we reached out to fathers through their spouses, which excluded men whom mothers did not want us to contact. However, as seen from the variability in scores on the Dyadic Adjustment Scale and given that all but 6 mothers who were asked agreed to father participation, this does not seem to have precluded families with poor-quality spousal relationships from being included in our sample. Second, some family-level data were collected from only the father or the mother. Third, given the challenges of sampling Syrian refugee mother-father-child triads in Jordan, the sample size – particularly for the fathers – is limited. This impacted our ability to test a range of covariates in the analyses that may be related to our outcomes. Fourth, data collection was undertaken at only two time points, which does not establish causality nor rule out other factors potentially impacting spousal agreement, family functioning, or mental health. This is complicated by the fact that the T1 and T2 mother/child and father data collection overlap but were not wholly synchronous. However, conducting both cross-sectional and longitudinal analyses increases confidence in the findings. Fifth, we have run numerous pre-registered analyses in this paper without correction for multiple testing; results should be interpreted in line with the increased possibility of Type I error. Sixth, some of the measures have quite low internal reliability (T1 mother-child closeness T2 mother-child inconsistent discipline, T2 internalising problems, T2 externalising problems). Finally, while our measure of father involvement is strong in that it was developed with an Arabic-speaking population in Lebanon, it does not specifically assess father involvement with the study child but rather assesses father involvement with all children in the household, whereas our other measures are specific to the study child. This picture of father involvement with their children may not be indicative of father involvement with each individual child.

Conclusion

This study helps to build evidence on father involvement among refugee father-mother-child triads living in adversity. Levels of father-mother disagreement with respect to fathering involvement had negative associations with relationship quality, maternal mental health, and child social and emotional learning. Our results suggest that helping parents negotiate caregiving and spousal interactions –

through education, child development, or social empowerment initiatives – could impact key aspects of family functioning, mental health, and child development.

Data availability

The data are in a data depository at: <https://osf.io/8bm6y>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1007/s10826-024-02809-y>.

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Compliance with Ethical Standards

Conflict of Interest The authors declare no competing interests.

Consent The fieldworkers provided participants with a consent form in Arabic, read it aloud to ensure full comprehension regardless of literacy, allowed time for reflection and addressed any questions or concerns, before obtaining informed consent. Mothers provided written consent for children to take part, after which children provided oral consent. They confirmed their willingness to have the father interviewed and provided his contact information. Fathers gave verbal consent, as they were interviewed by phone.

Ethics Approval The study was approved by the ethics committees of Yale University and Trinity College Dublin and by the Prime Minister's Office in Jordan. The fieldworkers provided participants with a consent form in Arabic, read it aloud to ensure full comprehension regardless of literacy, allowed time for reflection, and addressed any questions or concerns, before obtaining informed consent. Mothers provided written consent for children to take part, after which children provided oral consent. They were asked if they were willing to have the father interviewed and, if so, provided his contact information. Fathers gave verbal consent, as they were interviewed by phone.

Abbreviations

SEL social emotional learning

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